

Final Exam: Study Guide

Ethical Guidelines

1. A study recruits participants who are currently incarcerated. The researchers provide a brief overview of the study but fail to make clear that participation will not influence parole or sentence length. This scenario violates which principle of the Belmont Report?
2. Dr. Rivera is leading a study on a new medication and chooses only individuals from a low-income neighborhood to participate, without considering whether the benefits and burdens of the research are being fairly distributed. This is a breach of which ethical principle of the Belmont Report?
3. Anna, a student researcher, stores sensitive survey data on a shared drive without encryption or password protection. A classmate accesses the file accidentally while looking for another document. This situation violates which ethical principle outlined in the Belmont Report?
4. A research team collects blood samples from participants for a diabetes study. Later, they use the same samples for unrelated genetic testing without informing or getting additional consent from the participants. This is a breach of which ethical principle of the Belmont Report?
5. Michael is conducting interviews with participants about their experiences with mental health services. He forgets to inform them that participation is voluntary and that they can withdraw at any time. Some participants later feel obligated to continue despite feeling uncomfortable. This is a breach of which ethical principle of the Belmont Report?

Choose from this list:

- A. Beneficence
- B. Justice
- C. Respect for Persons

Research Paper

6. The main purpose of the Method section in an APA-style paper is to:
 - A. Describe how the study was conducted in enough detail to allow replication
 - B. Analyze the findings from the literature review
 - C. Interpret the results of the study

D. Discuss the broader implications of the study

7. In APA style, the abstract should:

- A. Be more than 500 words to provide full background
- B. Include only references and citations
- C. Provide a brief summary of the paper including the purpose, method, results, and conclusions
- D. Offer a detailed literature review

History of Knowledge

8. We often rely on authority figures for information. Which of the following is most important to consider when evaluating information from an authority figure?

- A. Whether they speak confidently
- B. Whether they are popular
- C. Whether they appear on television
- D. Whether they have relevant expertise and evidence to support their claims

9. Which of the following is an example of relying on authority for knowledge?

- A. Reading a peer-reviewed article
- B. Observing results from your own experiment
- C. Believing a doctor's advice about medication without reviewing the research yourself
- D. Conducting a survey and analyzing the data

10. What is one limitation of acquiring knowledge solely through authority?

- A. Authorities always rely on intuition
- B. Even experts can be wrong or biased
- C. Authority-based knowledge is rarely shared
- D. Authority-based knowledge is usually outdated

11. Maria believes her friend is upset because she just "has a feeling about it," even though her friend hasn't said anything. This is an example of using:

- A. Authority
- B. Logic
- C. Intuition
- D. Empiricism

12. Tom reads an article by a well-known psychologist and accepts the findings because the author is a leading expert in the field. Tom is relying on:

- A. Intuition
- B. Authority
- C. Empiricism
- D. Logic

13. Jasmine reasons that if all humans need water to survive, and Kevin is human, then Kevin must need water. This is an example of:

- A. Empiricism
- B. Intuition
- C. Authority
- D. Logic

14. Lena conducts an experiment where she observes how many times students raise their hands in class after different types of instruction. Her conclusions are based on what she directly observes. This is:

- A. Logic
- B. Empiricism
- C. Authority
- D. Intuition

15. "I just know this brand is better—my gut tells me." This statement best reflects:

- A. Empiricism
- B. Logic
- C. Intuition
- D. Authority

Experimental Designs

16. Chloe wants to study how adding morning meditation affects stress levels in college students. She finds a university that is starting an optional morning meditation program next semester. She collects data on student stress levels this semester and again next semester after the program begins. This study is best described as a:

- A. Experimental
- B. Quasi-experimental
- C. Non-experimental
- D. Longitudinal

17. Jamila is researching how installing a new reading program in a specific middle school affects students' reading comprehension scores. She cannot randomly assign students to use the new program but compares test scores from before and after the program was introduced. This is a(n):

- A. Experimental
- B. Quasi-experimental
- C. Non-experimental
- D. Longitudinal

18. A psychologist wants to study the relationship between sleep and academic performance. They survey 500 college students and ask about how many hours they sleep each night and their current GPA. What type of study is this?

- A. Experimental
- B. Quasi-experimental
- C. Non-experimental
- D. Longitudinal

19. Dr. Nguyen randomly assigns participants to two groups. One group is given a new memory-enhancing supplement, and the other group is given a placebo. Both groups then complete the same memory test. What type of study is this?

- A. Quasi-experimental
- B. Case study
- C. Experimental
- D. Non-experimental

20. Researchers are interested in how a new school policy affects student attendance. They compare attendance rates before and after the policy was implemented in one school, but they do not randomly assign students to conditions. What type of study is this?

- A. Experimental
- B. Non-experimental
- C. Observational

D. Quasi-experimental

21. A sociology researcher analyzes publicly available census data to explore the relationship between income level and access to public transportation in urban neighborhoods. What type of study is this?

- A. Quasi-experimental
- B. Experimental
- C. Non-experimental
- D. Mixed-method

22. A health researcher wants to test whether a new intervention reduces anxiety. Participants choose whether to receive the intervention or not, and the researcher compares anxiety levels afterward. What type of study is this?

- A. Experimental
- B. Quasi-experimental
- C. Non-experimental
- D. Correlational

Reliability & Validity

23. A researcher designs a scale to measure anxiety. Every time a participant takes the scale, they receive the same score. However, the scale mostly asks about physical fitness rather than emotional experiences. What does this scale demonstrate?

- A. Low validity, high reliability
- B. High validity, high reliability
- C. High validity, low reliability
- D. Low validity, low reliability

24. Which of the following best describes the concept of reliability?

- A. The consistency of a test or measurement tool over time
- B. Whether a test measures what it is supposed to measure
- C. The ability of a test to predict future performance
- D. The extent to which test results align with participant expectations

25. A bathroom scale gives you the same weight every time you step on it, but the weight is consistently 10 pounds too low. What does this suggest about the scale?

- A. It is both valid and reliable
- B. It is valid but not reliable
- C. It is reliable but not valid
- D. It is neither valid nor reliable

26. Which statement best defines validity?

- A. The extent to which a test can be repeated with the same results
- B. The degree to which a test accurately measures the concept it is intended to measure
- C. The correlation between two independent variables
- D. The variability of test results across different groups

Replication Crisis

27. What is one consequence of p-hacking in scientific research?

- A. It reduces the rate of true positives in research
- B. It enhances the reproducibility of research findings
- C. It increases the rate of false negatives in research
- D. It increases the rate of false positives in research

28. What does HARKing stand for in research practices?

- A. Hypothesizing after results are known
- B. Hypothesizing before results are known
- C. Hypothesizing during data collection
- D. Hypothesizing after data analysis

29. What is the main purpose of peer review in the publication process?

- A. To evaluate the quality and validity of research before publication
- B. To enhance research funding by improving grant application success
- C. To promote research visibility
- D. To expedite the publication process

30. How can researchers avoid questionable research practices?

- A. By selectively reporting only positive results
- B. By using flexible data analysis techniques

- C. By ensuring transparency, preregistration, and rigorous replication
- D. By relying solely on original study designs

Factorial Designs

31. What is a factorial design in research?

- A. An experiment where each level of one independent variable is combined with a subset of levels from other independent variables
- B. An experiment where each level of one independent variable is combined with every level of other independent variables
- C. An experiment where each level of one independent variable is combined with a random selection of levels from other independent variables
- D. An experiment where each level of one independent variable is combined with only one level of other independent variables

32. What is an interaction effect in the context of factorial designs?

- A. The additive effect of one IV, combining with the effects of other IVs
- B. The independent effect of one IV, regardless of the levels of other IVs
- C. The effect of one IV depends on the levels of another IV
- D. The overall effect of one IV, averaging across all other IVs

33. How many independent variables are involved in a 2 x 2 factorial design?

- A. One
- B. Two
- C. Three
- D. Four

34. In a mixed factorial design, what distinguishes between-subjects and within-subjects variables?

- A. Between-subjects variables use different groups of participants for each condition, while within-subjects variables use the same group for all conditions
- B. Between-subjects variables use independent groups for each condition, while within-subjects variables use independent groups for all conditions
- C. Between-subjects variables use the same group of participants for all conditions, while within-subjects variables use different groups for each condition

- D. Between-subjects variables involve manipulating all conditions for each participant, while within-subjects variables involve assigning different participants to each condition

35. Which one best explains the “main effects”?

- A. The effect of one IV that depends on the levels of another IV
- B. The effect of one IV only at a specific level of another IV
- C. The combined effect of all IVs without averaging across levels
- D. The overall effect of one IV, averaging across all other IVs

Recruitment & Survey Design

36. A researcher assigns a number to every student at a university and then uses a computer program to randomly select 100 students to participate in a survey. This is an example of:

- A. Cluster sampling
- B. Stratified sampling
- C. Simple random sampling
- D. Systematic sampling

37. A school district wants to study elementary school students, so the researcher randomly selects 5 schools and surveys every student in those schools. This is an example of:

- A. Cluster sampling
- B. Stratified sampling
- C. Systematic sampling
- D. Simple random sampling

38. A researcher selects every 10th name from a list of registered voters to participate in a study. This sampling method is known as:

- A. Simple random sampling
- B. Stratified sampling
- C. Cluster sampling
- D. Systematic sampling

39. A psychology student surveys people walking by in a shopping mall because it's quick and easy. This is an example of:

- A. Purposive sampling

- B. Quota sampling
- C. Snowball sampling
- D. Convenience sampling

40. A researcher studying underground music scenes begins by interviewing a few musicians, then asks them to refer other musicians to interview. This sampling method is:

- A. Snowball sampling
- B. Quota sampling
- C. Convenience sampling
- D. Simple random sampling

41. A marketing company needs a sample of 100 people, equally split between men and women. They go out and recruit people until they meet this target, but without using random selection. This is an example of:

- A. Snowball sampling
- B. Quota sampling
- C. Systematic sampling
- D. Cluster sampling

Hypothesis Testing

42. A researcher wants to test whether a new teaching method improves student test scores compared to the traditional method. What is the null hypothesis (H_0)?

- A. The new method improves test scores.
- B. There is no difference between the two teaching methods.
- C. The traditional method is worse.
- D. Students prefer the new method.

43. What does it mean if a researcher “fails to reject the null hypothesis”?

- A. The null hypothesis is definitely true.
- B. The alternative hypothesis is confirmed.
- C. There is insufficient evidence to support the alternative hypothesis.
- D. The results were statistically significant.

44. A researcher is testing a new drug and hypothesizes that it lowers blood pressure. Which of the following statements would be the alternative hypothesis?

- A. The drug has no effect on blood pressure.
- B. The drug lowers blood pressure.
- C. The drug raises blood pressure.
- D. Blood pressure stays the same regardless of the drug.

Choosing the right test

45. Which of these is **correctly** matched to its respective inferential test?

- A. Numerical IV & DV - Correlation
- B. Categorical DV & nominal IV - T tests
- C. Nominal IV & DV - Chi-square
- D. Numerical IV & categorical DV - ANOVAs

46. Short answer: Explain the difference between exploratory and planned tests.

Threats to Validity

47. What is internal validity in research?

- A. It is the extent to which the study's findings can be generalized to other populations or settings.
- B. It is the extent to which the study's statistical analyses accurately reflect the true relationship between variables.
- C. It is the extent to which the study can confidently conclude that the independent variable is the cause of changes in the dependent variable.
- D. It is the extent to which the study's measures consistently yield the same results over time.

48. Why might extreme scores become more average upon retesting?

- A. Extreme scores are due to consistent skill improvements that raise the average on retest
- B. Extreme scores are due to consistent environmental factors that maintain high or low performance on retest
- C. Extreme scores are due to consistent measurement errors that persist on retest
- D. Extreme scores are due to random fluctuations that regress toward the mean on retest

49. What does the term "order effect" refer to in pretest/posttest designs?

- A. It refers to the influence of external events on posttest scores

- B. It refers to the impact of pretest exposure on posttest performance
- C. It refers to the impact of selection bias on pretest scores
- D. It refers to the effect of random error on posttest scores

50. How might external events, like a pandemic, influence research outcomes?

- A. They may disrupt research outcomes by altering participants' behavior due to unexpected changes.
- B. They may enhance research outcomes by increasing participant engagement due to heightened awareness.
- C. They may have no effect on research outcomes as experimental controls remain unchanged.
- D. They may improve research outcomes by providing additional data points from diverse experiences.